

SONU SINGH

M.Tech – ThermoFluid Engineering | IIT Jodhpur

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EDUCATION

Degree	Institution	CGPA%	Year
M.Tech (TFE)	IIT Jodhpur	8.00 CGPA	2024
B.Tech (ME)	AKTU University	7.05 CGPA	2020
Senior Secondary	RSS International School	70.8%	2016
Secondary	Karl Huber International School	8.3 CGPA	2014

EXPERIENCE

Teaching Assistant | IIT Jodhpur, Jodhpur, India

Aug 2024 – Dec 2026

- MEL7060 – Heat Transfer Laboratory: Prepared and delivered tutorials, taught core heat transfer concepts, and graded assignments to support student learning.

Junior Research Fellow | IIT Jodhpur - IGSTC-WISER Project, Jodhpur, India

May 2026 - Aug 2026

(Ad hoc Position)

- Worked on an IGSTC-WISER funded project focused on high-temperature Phase Change Material (PCM)-based Thermal Energy Storage (TES) for industrial heat and Concentrated Solar Power (CSP) applications.
- Evaluated and selected high-temperature PCMs (300-1000 °C) based on thermophysical and energy-storage properties, and investigated PCM encapsulation for packed-bed TES systems.
- Performed numerical analysis of PCM melting and heat transfer, including temperature distribution and liquid fraction, to assess thermal storage performance.

Mechanical Engineering Intern | VaTech Escher Wyss Flovel Ltd, India

Jun 2019 – Aug 2019

- Conducted fluid dynamics simulations using ANSYS Fluent to improve efficiency in pumping and heat exchange systems.
- Assisted in the design and optimization of thermal systems for water treatment and power generation.

PROJECTS

CFD Analysis of Heat Transfer in PCM-Based Compact Heat Storage

Jan 2025 – May 2026

Master's Thesis | Supervisor: Dr. Shobhana Singh, Dept. of Mechanical Engineering, IITJ

- Performed transient CFD simulations to analyze melting and solidification in spherical encapsulated PCM for Thermal Energy Storage (TES).
- Executed meshing (structured/unstructured), Grid Independence Study, boundary conditions, solver settings (Pressure-based/Coupled), and Finite Volume Method (FVM).

- Post-processed temperature, liquid fraction, and non-dimensional numbers (Rayleigh and Nusselt); enhanced simulation accuracy with validation and verification.

Mechanical Characterization of Nano-Filled Polyester Composites

Sept 2019 – Apr 2020

Bachelor's Thesis | Supervisor: Dr. Akant Singh, Dept. of Mechanical Engineering, AKTU

- Investigated polyester composites reinforced with nano-fillers (silica and carbon nanotubes).
- Conducted tensile, flexural, and toughness tests; analyzed the effect of nano-filler concentration on strength and toughness.

TECHNICAL SKILLS

Programming: MATLAB, Python

Simulation & Design Software: ANSYS Fluent, COMSOL Multiphysics, SolidWorks, AutoCAD

POSITIONS OF RESPONSIBILITY

- Placement Coordinator, M.Tech Thermofluid Engineering, IIT Jodhpur (2025–26)
- Organizing Team Member, X-SEEC Conference, IIT Jodhpur – Assisted in event coordination, logistics, and team support.
- Organizing Team Member, ISHMT Conference, IIT Jodhpur – Assisted in event coordination, logistics, and team support.

PUBLICATIONS

- **Singh, S., Singh, S., Singh, S., & Niedermeier, K.** *MCDA-Guided Selection of PCM for High-Temperature Energy Storage Application and Numerical Assessment of Top-Ranked PCMs.* Proceedings of the ASME 2026 International Mechanical Engineering Congress and Exposition (IMECE-INDIA2026), Paper No. IMECE-INDIA2026-189225, September 03–05, 2026, India.
- **Singh, S., Singh, S., Kansara, K., & Niedermeier, K.** *Data-Driven Multi-Criteria Selection of Phase Change Materials for High-Temperature Energy Storage Application.* Book Chapter, Springer.

ACHIEVEMENTS

- Qualified GATE 2024 – Mechanical Engineering (ME).
- Secured 1st position in Techtrix Competition, North Region (2017).
- Achieved 1st position in Techno Competition, Mangalmay Group of Institutions (2017).